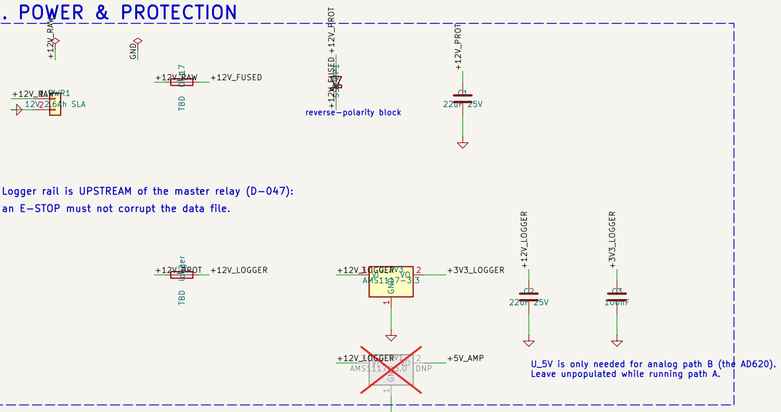
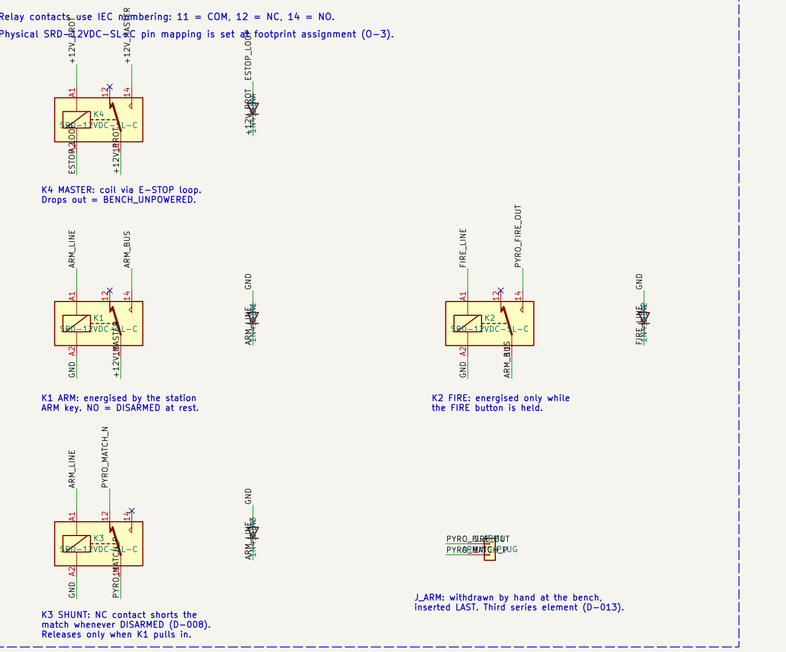
1. POWER & PROTECTION



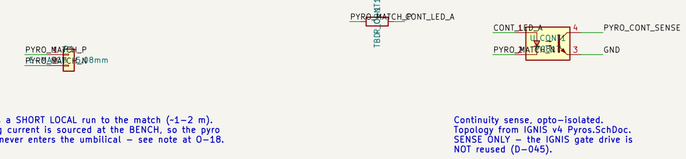
2. SAFETY CHAIN — hardware only, no MCU

Relay contacts use IEC numbering: 11 = COM, 12 = NC, 14 = NO.
Physical SRD-12VDC-SL-C pin mapping is set as footprint assignment (D-3).



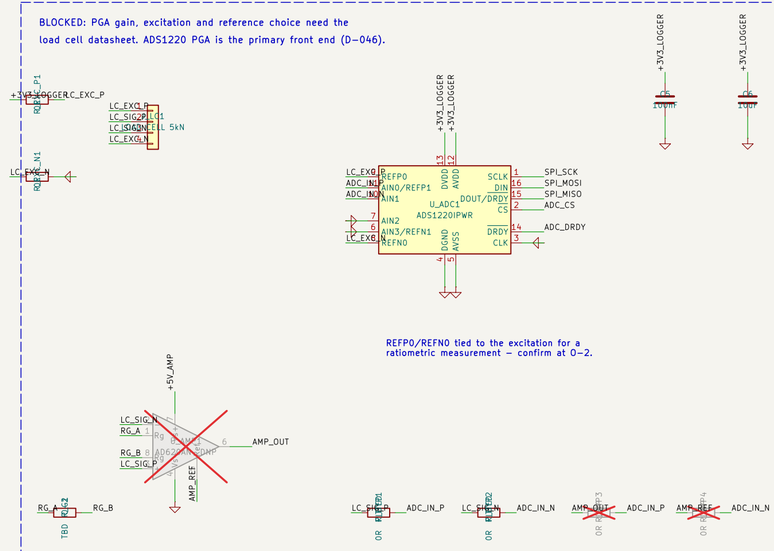
3. PYRO INTERFACE — values blocked on O-1

BLOCKED: R_CONT, fuse and relay contact ratings all need the
e-match no-fire current. Topology is complete; values are TBD.



4. ANALOG FRONT END — values blocked on O-2

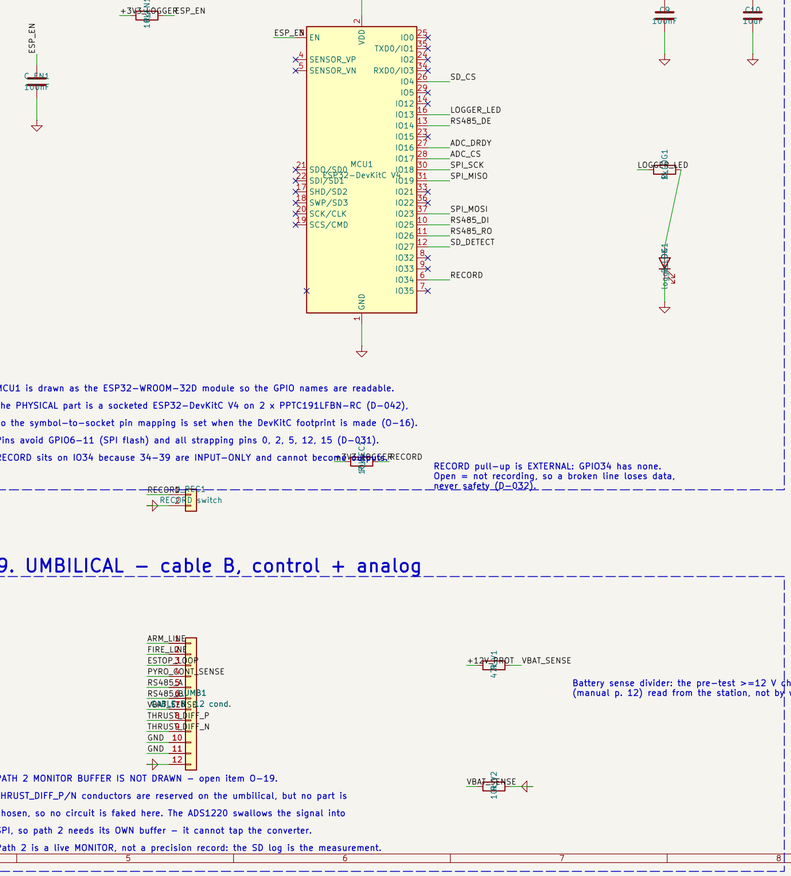
BLOCKED: PGA gain, excitation and reference choice need the
load cell datasheet. ADS1220 PGA is the primary front end (D-046).



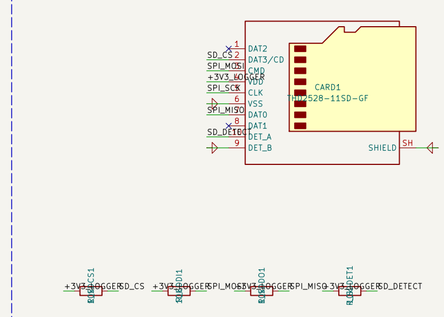
AD620 minimum total supply is 4.6 V, so it CANNOT run on 3.3 V.
Path B therefore needs the +5V_AMP rail, and its output must be kept
inside the ADS1220 input range (0-7). Single-supply use also needs
AMP_REF biased above ground — value TBD with O-2.

5. LOGGER — ESP32-DevKitC V4, zero authority

No outputs reach any safety net. Radios stopped at boot (D-029).

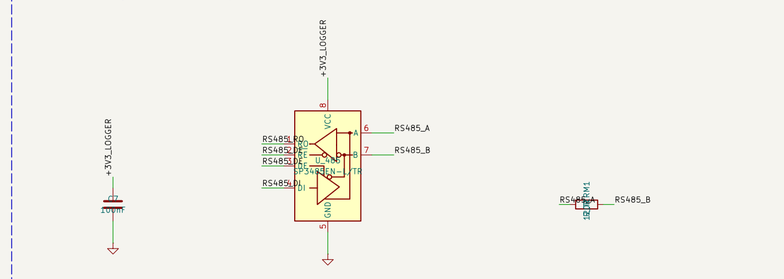


6. STORAGE — on-board microSD



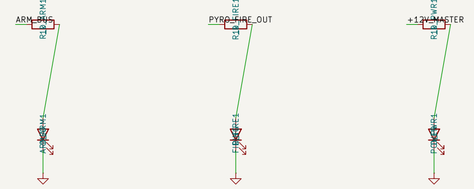
7. TELEMETRY — isolated RS-485 to the station

Data OUTWARD only. Never a command path.



8. INDICATORS — beacon driven from the REAL arm/fire lines

D-017: the beacon must not be able to misreport state,
so it is NOT driven by the MCU. WS2812B cannot serve here (D-043).



9. UMBILICAL — cable B, control + analog

TBD values await O-1 (e-match no-fire current) and O-2 (load cell datasheet).

Review owner: Emilio Guadarrama. Generated by fabrication/generate_schematics.py
Igniter safe state is SHORTED via K3 normally-closed. Values marked TBD await O-1 / O-2.
PRELIMINARY — NOT APPROVED. No MCU on the arm, fire or e-stop path.

Ignitia Rocket Lab

Sheet: /
File: talavera-bench.kicad_sch

Title: TALAVERA v1 — Bench Board

Size: A2	Date: 2026-11-06	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 1/1